

Dilshani Karunarathna

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SUMMARY

Software Engineer with full-stack development and delivery experience across React, TypeScript, Node.js, C#, .NET, and cloud platforms. Experienced in owning features end-to-end, from requirements and technical design through frontend and backend development, testing, deployment, and production support. Skilled in building scalable application features, integrating REST APIs, improving performance, and developing automated tests to support reliable software delivery.

EXPERIENCE

IFS R&D International (Pvt) Ltd (Global org, 7,000+ employees) | Colombo, Sri Lanka

Jul 2025 – Jul 2026

Senior Software Engineer

- Coordinated release engineering for two major enterprise releases and multiple monthly releases across 10+ engineering teams and cloud environments, overseeing deployment, validation, post-release verification, and production support.
- Reduced documentation deployment time by 98% (4 hours to under 3 minutes) by architecting an automated CI/CD documentation pipeline using Azure Static Web Apps and MkDocs, now adopted as the team's standard release-documentation workflow.
- Improved release quality by strengthening deployment pipelines, increasing unit test coverage using SonarQube, remediating security vulnerabilities, and enforcing code quality standards across enterprise applications.
- Performed root cause analysis across cloud infrastructure, CI/CD pipelines, and enterprise applications, implementing long-term solutions that improved platform reliability and operational resilience.
- Designed and delivered an AI-powered engineering assistant using Microsoft Copilot Studio that indexed over 100 internal knowledge pages, increasing documentation usage fivefold and reducing time spent locating technical information.
- Coordinated solution design and technical decision-making, evaluating implementation options, mitigating technical risks, and delivering scalable, secure, and maintainable enterprise software.
- Mentored junior engineers through code reviews, knowledge sharing, and technical guidance while promoting engineering best practices across the team.
- Applied AI-assisted coding tools, including GitHub Copilot and Cursor, to accelerate implementation, debugging, refactoring, test generation, and documentation while maintaining code quality, security, and engineering standards.

Xyicon LK (Pvt) Ltd | Colombo, Sri Lanka

Sep 2022 – Jul 2025

Software Engineer

- Delivered more than 10 production features across frontend and backend systems, owning development from requirements analysis and technical design through implementation, testing, deployment, and production support while collaborating with product owners, UI/UX designers, QA engineers, and other development teams.
- Developed scalable web application features using React, TypeScript, and Node.js, contributing across the full application stack and integrating frontend functionality with backend services and REST APIs.
- Developed and enhanced backend services using C# and .NET, implementing REST APIs, business logic, data processing, database interactions, and cloud storage integrations to support scalable end-to-end application features.
- Improved application performance and maintainability by implementing caching strategies, optimizing data retrieval, reducing redundant API calls and client-side memory usage, resulting in faster and more responsive user workflows.
- Contributed to the Revit add-in using WPF and XAML, integrating desktop engineering workflows with cloud-based applications.
- Resolved over 100 software defects and enhancement requests, improving platform stability, maintainability, and user experience.
- Cut manual regression testing time by 89% (3 hours 20 minutes) by building over 100 automated end-to-end tests and more than 20 unit tests.
- Participated in architecture and technical design discussions, code reviews, technical planning sessions, and Agile ceremonies to continuously improve software quality and engineering standards.

Temporary Instructor

- Published peer-reviewed research in IEEE Access by extending undergraduate research into a multidisciplinary engineering publication.
- Supported faculty research through software development, testing, debugging, technical documentation, and data analysis.
- Mentored undergraduate engineering teams by providing technical guidance, coordinating Agile development activities, and supporting successful project delivery.

Geveo Australasia Pvt Ltd | Colombo, Sri Lanka

Apr 2021 – Aug 2021

Software Engineer Intern

- Actively contributed to two full-stack software development projects within Agile teams.
- Developed and enhanced application features in alignment with business requirements and functional specifications, ensuring high performance, reliability, and scalability.
- Diagnosed and resolved functional and performance issues, and implemented unit tests to maintain code quality, stability, and long-term maintainability.

EDUCATION

Bachelor of Science of Engineering (First Class Honours) | University of Peradeniya Nov 2016 – Sep 2021

Computer Engineering, GPA: 3.70/4.00

SKILLS

Programming Languages: TypeScript, JavaScript, C#, Python, Java, SQL

Frontend Development: React, React Hooks, jQuery, HTML5, CSS3

Backend Development: Node.js, REST APIs, .NET

Cloud & DevOps: Microsoft Azure, Azure DevOps, Azure Functions, Azure Static Web Apps, Azure Storage, Azure CLI, Azure Resource Manager (ARM), Microsoft Entra ID, CI/CD, GitLab, Bitbucket Pipelines

Testing & Quality Assurance: Cypress, Unit Testing, End-to-End Testing, Test Automation, SonarQube

Automation & Documentation: Microsoft Copilot Studio, Microsoft 365 Copilot, PowerShell, Bash, YAML, JSON

Development Tools & Frameworks: Git, Visual Studio, Visual Studio Code, Postman, WPF, MkDocs

Engineering Practices: Object-Oriented Programming (OOP), Design Patterns, Release Engineering, Agile, Scrum, Root Cause Analysis, Production Support, Technical Documentation

RESEARCH PUBLICATIONS

Mixed Reality-based Multi-Agent Robotics Framework for Artificial Swarm Intelligence Experiments

IEEE Access: The Multidisciplinary Open Access Journal DOI: [10.1109/ACCESS.2023.3317434](https://doi.org/10.1109/ACCESS.2023.3317434)

Segmentation and significance of herniation measurement using Lumbar Intervertebral Discs from the Axial View

Moratuwa Engineering Research Conference DOI: [10.1109/MERCon55799.2022.9906245](https://doi.org/10.1109/MERCon55799.2022.9906245)

ACCOMPLISHMENTS

- **MERCon 2021:** Best Research Paper
- **ESCaPe 2021:** Best Research Presentation
- **ACES Hackathon 2019:** First Place
- **IEEEExtreme 13.0:** Country Rank: 21, Global Rank: 436
- **IEEEExtreme 12.0:** Country Rank: 38, Global Rank: 499
- **Google Hash Code 2019:** Country Rank: 24

CERTIFICATIONS

AWS Fundamentals: Going Cloud-Native, Amazon Web Services (ID: BT7ULEWL3M67)	2020
Google Cloud Platform Fundamentals: Core Infrastructure, Google Cloud (ID: 2K6MB4MJ4KX4)	2020
Neural Networks and Deep Learning, DeepLearning.AI (ID: R5RM43YJPFT2)	2020
Introduction to Big Data and Hadoop, BdREN (Country Rank 2, Global Rank 9) (ID: fDLuDCf-DLE4-2132)	2020